

Chaitanya Devarshi

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TECHNICAL SKILLS

Languages and APIs: Python, SQL, OpenAI API, LangChain, REST API

Data Handling & Retrieval: Vector Database, Data Wrangling, ETL Pipelines

ML/DL & GenAI: PyTorch, Scikit-learn, Hugging Face, Fine Tuning LLMs, AutoGen, RAG Pipelines, Transformers, NLP

Visualisation and Storytelling: Tableau, Matplotlib, Seaborn, Excel, EDA

MLOps and Testing: PyTest, GitHub Actions, Prompt Evaluation

Tools & Platforms: Git, JIRA, Azure, AWS, Weights and Biases, Streamlit

WORK EXPERIENCE

Persistent Systems Ltd.

Sep 2021 - Dec 2023 | Pune, India

Senior Software Engineer | Client - *Optum UnitedHealth Group*

Jul 2023 - Dec 2023

- Directed the integration of RAG pipelines into existing data workflows using LangChain and FAISS, reducing knowledge retrieval latency by 40% for GenAI agents.
- Formulated Python-based ETL and vectorization scripts with Hugging Face Transformers to generate domain-specific embeddings for healthcare documents, enabling context-rich LLM interactions.

Software Engineer | Client - *IQVIA*

Sep 2021 - Jun 2023

- Crafted transformer-based NLP models using PyTorch and Hugging Face to summarize clinical notes, achieving a 92% ROUGE-L score, and significantly enhancing medical insights extraction.
- Streamlined LLM fine-tuning workflows for patient data summarization using LoRA-based tuning techniques, reducing model deployment costs by 30%.
- Supervised a 7-member team in deploying OpenAI API-based GenAI tools for patient query resolution and dashboard Q&A interfaces, improving user engagement metrics by 25%.

PROJECTS

E-commerce Customer Segmentation and Purchase Prediction | *Python, Clustering, KNN Modeling*

- Implemented a KNN-based segmentation model on e-commerce data with 97.13% accuracy, identifying three key customer groups to drive data-backed personalization and marketing strategy.
- Refined 12 key behavioral and demographic features using feature engineering techniques to improve segmentation effectiveness and increase customer conversion rates through targeted analytics insights.

High-Risk Heart Disease Prediction Model | *Python, SVM, Neural Networks, Random Forest*

- Investigated heart disease data from 1.3 million patients to identify individuals with a high probability of future heart attacks.
- Constructed machine learning models (Random Forest, SVM) to predict high-risk patients, achieving a prediction accuracy of 92%.

New York Motor Vehicle Collisions - Crashes | *Tableau, Storytelling*

- Analyzed NYPD collision datasets using Tableau to identify high-risk boroughs and seasonal spikes, revealing a 30% increase in incidents during July and August in Brooklyn.
- Visualised NYPD collision data in an interactive Tableau dashboard, pinpointing high-risk zones; analysis informed resource allocation and reduced response times in critical areas by 15%.

Disease Data Modeling and Warehousing Project | *PostgreSQL, ETL, Data Modeling, AWS, DbSchema*

- Devised dimensional data models, ER diagrams, and interactive dashboards to visualize disease trends and evaluate data warehousing solutions, resulting in a 20% improvement in cost performance through AWS optimization.
- Engineered an ETL pipeline and scalable data warehouse using PostgreSQL to streamline disease data integration and improve analytics processing efficiency by 30%.

EDUCATION

Katz School of Science & Health - Yeshiva University

Jan 2024 – Expected May 2025 | New York, NY

Master of Science in Data Analytics & Visualization

Cumulative GPA: 3.55

Sinhgad College of Engineering

Aug 2017 – July 2021 | Pune, INDIA

Bachelor of Engineering in Electronics & Telecommunication

GPA: 3.2

CERTIFICATIONS, AWARDS & INTERESTS

- Certifications:** Advance Certificate Programme in Data Science | *IIIT Bangalore*, Data Science & Machine Learning Complete Course | *Coding Ninjas*
- Awards:** National Award from HRD Minister for creating Sanskrit Learning Games, Top Talent 2023(Persistent Systems Ltd.), Played Nationals in Basketball & Volleyball.
- Interests:** Swimming, Soccer, Volleyball, Basketball, Table Tennis